KCE PATHWAYS · GRADES 4–5 · UNIT 5: CHARTING THE COURSE · WEEK 21 · MATH

The Yellowstone Trail Map

Name: _____ **Date:** _____

PRE-LESSON WARM-UP · DO THIS BEFORE YOUR COMPUTER LESSON

Pre-Lesson Warm-Up: Finding Your Way

Quinlan is spending a week in Yellowstone National Park with his cousin Ramona. Warm up before the lesson and show your work. Some problems may feel new, and that is okay!

1. Review (Unit 4, volume): a) A storage bin is 7 feet long, 5 feet wide, and 2 feet tall. What is its volume? b) A box holds 240 cubic inches. Its base is 10 inches long and 6 inches wide. How tall is it?

2. Start at 0 and count by 4s. Write the first five numbers. Then start at 0 and count by 8s. Write the first five numbers.

3. Look at your two lists. Compare the numbers in the same position (the 2nd and the 2nd, the 3rd and the 3rd). What do you notice?

4. A trail map says the campsite is at (3, 5). What do you think the two numbers tell you to do?

   Move 3 right, then 5 up | Move 3 up, then 5 right | Move 8 steps in any direction

5. Draw two number lines from 0 to 6 that meet at 0, one going right and one going up. Mark a dot 2 steps right and 4 steps up.

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POST-LESSON CHALLENGE · DO THIS AFTER YOUR COMPUTER LESSON

Post-Lesson Challenge: Ranger Roderick's Map

Walk right for the x-coordinate, then climb up for the y-coordinate. Show your work.

1. a) Write the ordered pair for a point 7 units right of the origin and 2 units up. b) Is the point (0, 6) on the x-axis or the y-axis? c) Start at (5, 4). Move 3 units left and 4 units up. Where do you stop?

2. Two steps: the Ranger Station is at (2, 3) and the Overlook is at (2, 9). Each grid unit stands for 1/4 mile. How many miles apart are they? How many miles is a trip there and back?

3. Rule A: start at 0, add 6. Rule B: start at 0, add 3. Write the first 5 numbers of each rule, then the ordered pairs (Rule A, Rule B). What is the relationship?

4. Ramona graphs (hours hiked, miles walked): (1, 3), (2, 6), (3, 9). If the pattern continues, which point shows 7 hours?

   (7, 21) (21, 7) (7, 12)

5. Find the mistake: Ruben says (4, 1) and (1, 4) are the same point because they use the same numbers. Explain his mistake in two or three complete sentences.

★ **Challenge:** Rule A: start at 1, add 2. Rule B: start at 4, add 2. Write four ordered pairs. Is each Rule B number "3 times" or "3 more" than Rule A? How do you know?

Looking Ahead: Week 22: Ship Day in Duluth Harbor. Convert customary and metric units and solve multistep problems.